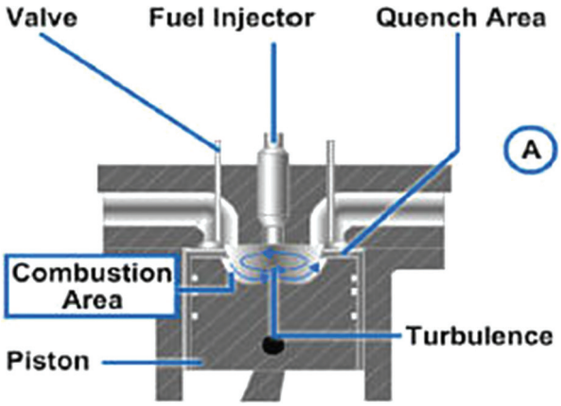


Figure 6.11.1(a) Open Combustion Chamber

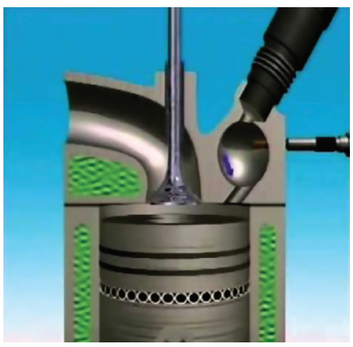


Figure 6.11.1(b) Pre-Combustion Chamber

The mixture first starts ignited in the auxiliary chamber and the combustion flame travels towards the main combustion chamber for the burning the rest of the air fuel mixture. Normally Glow plug is used to initiate combustion of the mixture. The glow plug will be located at the auxiliary chamber. Figure 6.11.1(b) shows the Pre-Combustion Chamber.

(c) Swirl Combustion chamber

It is also known as the turbulent chamber. The air enters into the combustion chamber in swirl motion in which the fuel is injected on the swirl air motion, allowed to mix with the air and achieves

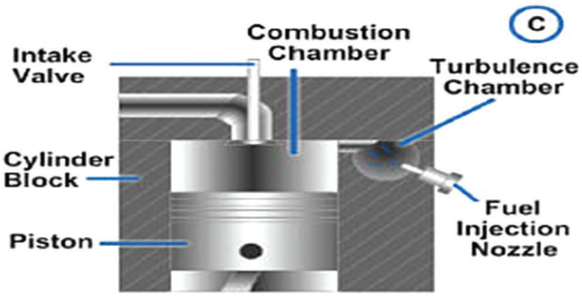


Figure 6.11.1(c) Swirl Combustion Chamber

combustion. Figure 6.11.1(c) shows the Swirl Combustion Chamber.

(d) Squish Combustion chamber

The piston head has the bowl shape in which the air motion is travelled from the side to centre of chamber. The radically inward movement of the air is called as squish. As the piston moves from BDC to TDC, the squish motion is created and the fuel is injected and ignited. Figure 6.11.1(d) shows the Squish Combustion Chamber.

(e) Air cell and energy cell

Air cell

The Air cell design has two (called as air cell and main) chambers in which they

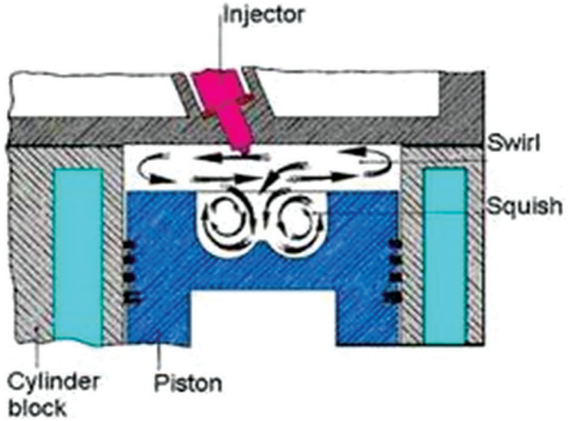


Figure 6.11.1(d) Squish Combustion chamber

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